

Cotranslational membrane protein biogenesis by an EMC-bound translocon

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Secretory and membrane protein biogenesis occurs at ribosome-bound translocons in the endoplasmic reticulum (ER) membrane. Translocons consist of the Sec61 protein-conducting channel and various maturation factors including chaperones, modification enzymes and membrane insertases, which assemble dynamically in response to the biogenesis needs of a nascent polypeptide. In human cells, nascent multipass membrane proteins utilize the ER membrane protein complex (EMC) and the multipass translocon (MPT), but how these factors are deployed during biogenesis is unknown. Here we combine selective ribosome profiling, cryo-electron tomography and site-specific photocrosslinking to reveal the substrate range, timing and mechanism of EMC recruitment. We show that multipass proteins initially engage the MPT for insertion and chaperoning, with EMC being recruited when substrates begin to exceed the cytosolic or intramembrane capacity of MPT. Substrate-engaged EMC is observed at the interface between the MPT's protected membrane cavity and the bulk ER membrane, where it is poised for insertion, chaperoning and assembly. These results show how EMC and MPT cooperate at the ribosome-bound translocon to facilitate multipass membrane protein biogenesis.

Human cells encode thousands of integral membrane proteins with diverse topologies and architectures. Most of these are cotranslationally inserted into the endoplasmic reticulum (ER) membrane at translocons whose components coordinate nascent chain translocation, membrane integration, modification and folding¹. ER translocons are centered around the Sec61 protein-conducting channel, which binds the ribosome and can direct the nascent chain into or across the lipid bilayer^{2,3}. It was long thought that all membrane protein insertion occurred through a lateral gate in the Sec61 channel^{4–6}. However, potent lateral gate inhibitors^{7–9} have recently revealed that many, if not

most, multipass membrane proteins are inserted by non-Sec61 routes facilitated by members of the universally conserved Oxa1 superfamily of insertases^{10–12}.

At the mammalian ER, two abundant Oxa1 superfamily members facilitate multipass protein biogenesis. The first is the ER membrane protein complex (EMC), a large nine-subunit complex^{13–15} best characterized as an insertase for transmembrane domains (TMDs) near the beginning or end of a protein^{16–19}. The second is GEL, a two-subunit complex that is proposed to facilitate cotranslational insertion of pairs of TMDs within multipass proteins. The GEL, PAT and BOS complexes

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(collectively called MPT) are recruited to Sec61-bound ribosomes to form the multipass translocon^{11,12,20–22}. Unlike Sec61 inhibitors, loss of EMC or MPT components impairs a wide range of multipass membrane proteins to varying degrees, implicating both of these factors in their biogenesis^{7,11,17,20,23–25}.

In addition to TMD insertion, both EMC and MPT probably chaperone membrane proteins to protect them from promiscuous degradation, facilitate their folding and aid in their assembly with other proteins^{11,20–22,24,26–29}. Thus, the topologically diverse set of ~2,500 human multipass membrane proteins exploit the activities of Sec61, EMC and MPT at different times and in different ways to ensure their successful maturation. The rules by which these and other factors are deployed are only beginning to emerge. Here we focus on EMC, whose cotranslational role in membrane protein biogenesis has been hinted at in early proximity-labeling studies²⁴, but whose substrate range, recruitment mechanism and coordination with MPT all remain unclear.

Results

EMC can engage MPT-bound ribosomes during multipass synthesis

A subpopulation of EMC was found to cofractionate through a sucrose gradient with Sec61-associated ribosomes prepared from nuclease-treated, digitonin-solubilized human embryonic kidney 293 (HEK293) membranes (Extended Data Fig. 1). To identify the proteins being synthesized on EMC-engaged ribosomes, we affinity purified these complexes via tagged EMC subunits and sequenced the ribosome-protected messenger RNA (mRNA) fragments^{30,31}. The EMC-selective sequencing data, obtained using HEK293 cell lines stably expressing Flag-tagged EMC6, EMC7 and EMC10, were compared to matched ribosome-protected mRNA sequences obtained from total membranes.

As expected for subunits of the same complex, transcript enrichment was highly correlated for EMC6, EMC7 and EMC10, allowing us to combine the datasets into a single sample ($n = 7$), hereafter referred to as 'EMC' (Extended Data Fig. 2). Of the ~5,800 proteins detected in the combined EMC dataset, multipass proteins were the most enriched class, with 38% of type I, 62% of type II and 57% of type III multipass proteins showing $\log_2$ enrichment ≥ 2 (Fig. 1a). In contrast to MPT, shown previously to enrich most multipass proteins regardless of TMD number¹², EMC recovered only a subset, with enrichment increasing with the number of TMDs (Fig. 1b,c).

Comparison of the EMC and MPT datasets showed that about half of multipass transcripts were enriched by both EMC and MPT, about one-third by only MPT and essentially none by only EMC (Fig. 1d). Considered with the finding that EMC favors multipass proteins with more TMDs than MPT (Fig. 1b,c), the data indicate that EMC is recruited to MPT-engaged ribosomes. This is illustrated by ABCD4, a 6-TMD type II ABC transporter, which recruits MPT as its first two TMDs emerge from the ribosome, followed by EMC as the next pair of TMDs emerges (Fig. 1e). Notably, MPT does not depart from the substrate on EMC recruitment.

To validate this sequential recruitment to the same ribosome, we analyzed the translocon composition of ABCD4 intermediates assembled by *in vitro* translation. Whereas a 2-TMD intermediate recovered MPT but not EMC, a 4-TMD intermediate recovered both and neither recovered the oligosaccharyltransferase complex (OST-A), a component of the secretory translocon (Fig. 1f). A second purification step of the 4-TMD intermediate via tagged EMC10 recovered MPT, showing that both EMC and MPT are coassembled with ABCD4 at this stage (Fig. 1g,h). Similar results were seen if tandem affinity purification was performed via ABCD4 followed by an MPT component, but not by an OST-A component (Extended Data Fig. 3a). Thus, EMC can engage MPT-bound ribosomes during multipass protein synthesis.

Substrate-directed EMC recruitment to the multipass translocon

To understand how EMC is recruited to the ribosome, we analyzed the mRNA positions enriched in EMC-associated ribosomes compared with those enriched by MPT. This positional analysis revealed that EMC typically arrived after MPT, and once engaged, persisted until synthesis was complete (Fig. 2a). This differs from MPT, whose binding can be repeatedly and reversibly disrupted during synthesis of long cytosolic and luminal segments¹². An example is POMT1, an 11-TMD type II protein, which recruited MPT early followed later by EMC. Synthesis of a long downstream luminal segment, which would need to be translocated through Sec61, resulted in disengagement of MPT and recruitment of OST-A, but had little effect on EMC binding (Fig. 2b). Similarly, despite repeated cycles of MPT binding and disengagement during synthesis of very long cytosolic segments in the 10-TMD type II protein ATP11A, EMC remained engaged throughout (Fig. 2c). Thus, EMC is preferentially recruited to ribosomes containing MPT but, once engaged, it can remain bound regardless of any downstream MPT and OST-A dynamics.

EMC arrived at the translocon when between two and eight TMDs had emerged from the ribosome (Fig. 2d). This broad distribution contrasts with MPT, which typically arrives at closed Sec61 as the first TMD or closely spaced TMD pair begins to emerge from the ribosome¹². Previous structural analyses suggested that about 6–8 TMDs can be accommodated within the MPT central membrane cavity^{11,21} (Fig. 2e, left). This coincides with the upper limit of EMC onset, suggesting that EMC can be recruited when the nascent chain TMDs begin to exceed MPT's intramembrane capacity.

We noted that multipass proteins with many cytosolic residues early in their sequence often recruited EMC early, when few TMDs were inserted. An example is given by ANO5, a 10-TMD type II protein with a 301-residue cytosolic N-tail, which engaged MPT and EMC after its first (closely spaced) TMD pair emerged from the ribosome (Extended Data Fig. 3b). Conversely, proteins with few cytosolic residues early in their sequence often recruited EMC later. An example is given by SLC30A5, a 16-TMD type II protein with short cytosolic segments that engaged MPT early but did not recruit EMC until TMD6 had emerged from the ribosome (Extended Data Fig. 3c).

To quantify this, we classified proteins by the total length of their first two cytosolic segments and plotted the cumulative fraction engaged with EMC as a function of the number of TMDs emerged at onset (Fig. 2f). The midpoint progressively shifted from 5.1 TMDs for proteins with the fewest cytosolic residues (0–30 and 31–60 residues), to 3.8–4.4 TMDs for proteins with an intermediate number (61–120 residues), to 2.1 TMDs for proteins with the most (>120 residues). This effect was specific to cytosolic residues, because the timing of EMC onset was largely agnostic to the length of the first two luminal segments (Fig. 2g).

The cytosolic vestibule between the ribosome exit tunnel and the central membrane cavity is small and largely shielded from the surrounding cytosol (Fig. 2e, right). Depending on the conformation of the nascent chain (and excluding smaller side cavities), we estimate that about 60–90 residues can be accommodated within the vestibule. Consistent with this, the median number of cytosolic residues emerged at EMC onset (or before TMD6, whichever occurs first) was 142 for early onset proteins ($\leq$ TMD3) and 67 for late onset proteins ($\geq$ TMD5) (Fig. 2h). This suggests that EMC can be recruited when the nascent chain can no longer be accommodated within the MPT cytosolic vestibule.

Thus, the timing of EMC onset is influenced by at least two properties of the elongating nascent chain: the number of TMDs and cytosolic residues emerged prior to onset. For multipass proteins with fewer than about six TMDs, EMC onset is favored by the emergence of many cytosolic residues, whereas those exposing few cytosolic residues recruit EMC weakly, or not at all (Extended Data Fig. 3d,e). For proteins with more than about six TMDs, early EMC recruitment is also favored by the emergence

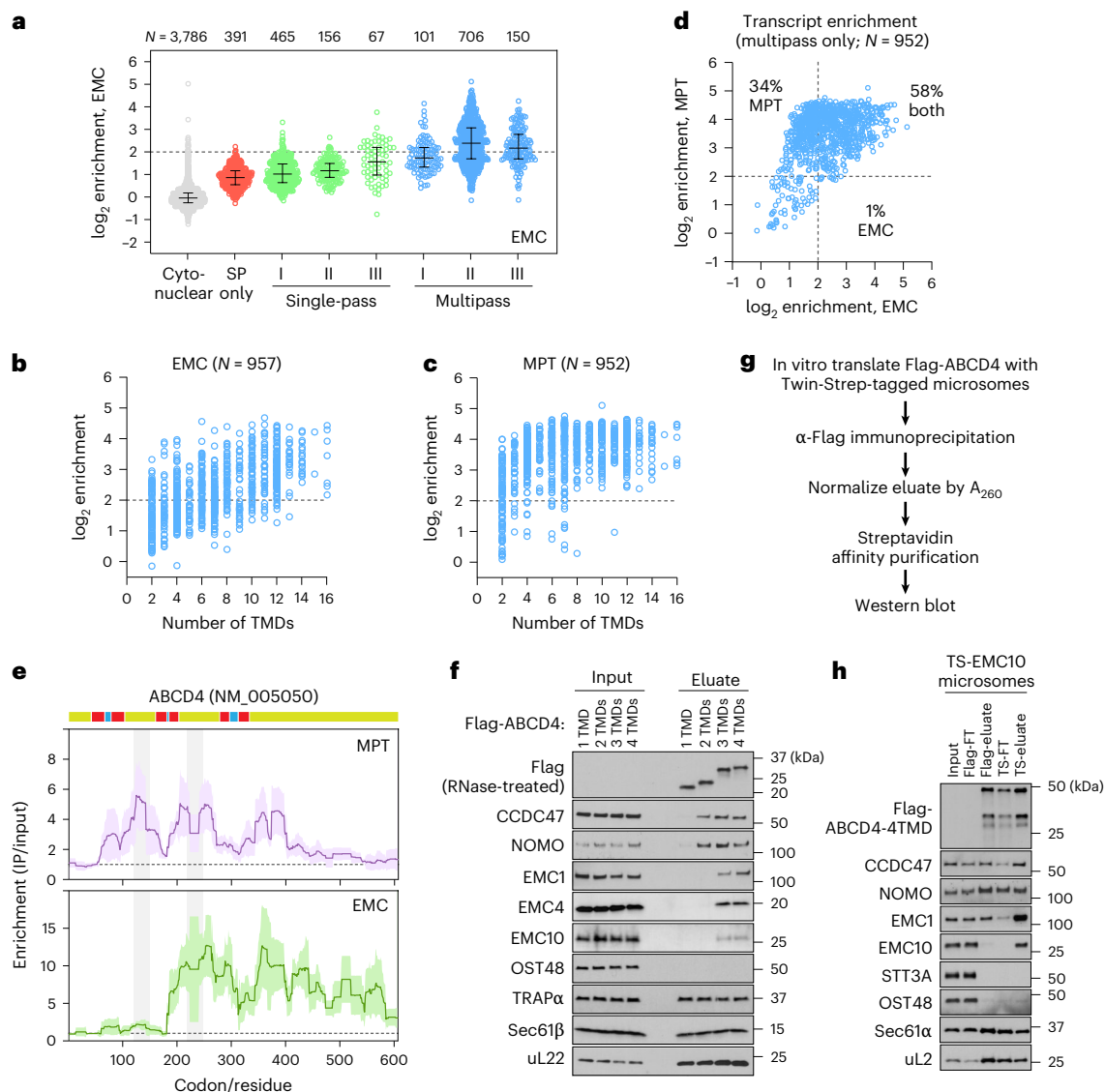


Fig. 1 | Selective ribosome profiling identifies cotranslational substrates of EMC. **a**, EMC $\log_2$ enrichment of transcripts encoding proteins of the indicated type. Enrichment is calculated as the average of EMC6 (N = 2), EMC7 (N = 2) and EMC10 (N = 3) samples. The number of sequences (N), median and interquartile range are indicated. **b**, Scatterplots of $\log_2$ transcript enrichment versus the number of TMDs for multipass proteins detected in the EMC sample. **c**, As in **b**, but for a previously reported MPT sample¹². **d**, The $\log_2$ enrichment scatterplot for 952 multipass-encoding transcripts detected in EMC and MPT samples. **e**, EMC interaction profile for ABCD4. Mean enrichment (line) and standard deviation (shaded) for $n = 7$ replicates are indicated. A diagram of ABCD4 is shown above, indicating cytosolic (yellow), transmembrane (red) and luminal

(blue) segments. Residues to the left of the vertical gray bars have emerged from the ribosome exit tunnel in the 2-TMD and 4-TMD constructs characterized in **f**. **f**, Stalled, Flag-tagged ABCD4 constructs truncated 45 residues downstream of the indicated TMD were translated in rabbit reticulocyte lysate in the presence of HEK293-derived membranes, and the membrane-associated fraction was isolated by sedimentation. Following anti-Flag affinity purification of the digitonin-solubilized membranes, the translocon composition was analyzed by SDS-PAGE and immunoblotting. **g**, Strategy for tandem affinity purification of the Flag-tagged 4-TMD intermediate after translation in the presence of Twin-Strep-tagged EMC10 membranes. **h**, Analysis by SDS-PAGE and immunoblotting. SP, signal peptide. IP: immunoprecipitation.

of many cytosolic residues, whereas those exposing few cytosolic residues recruit EMC later, after about 6–8 TMDs have emerged from the ribosome. These observations are consistent with a simple model in which EMC is recruited to the translocon when the growing nascent chain begins to exceed the cytosolic or intramembrane capacity of MPT.

Architecture of the EMC-bound multipass translocon

To visualize how EMC is organized at the translocon during multipass synthesis we programmed HEK293 membranes with the 4-TMD ABCD4 intermediate and subjected these to cryo-electron tomography (cryo-ET) and subtomogram averaging (Table 1 and Extended Data Fig. 4). Several classes were observed, corresponding

to core, secretory and multipass translocons³². The multipass translocon class could be further classified into two populations (Fig. 3a). The first, with an overall resolution of ~ 9.7 Å, showed clear density for the Sec61, GEL, PAT and BOS complexes, and weaker density for the dynamic TRAP complex. The second, with an overall resolution of ~ 21.8 Å, was similar except for a prominent elongated density spanning the membrane near Sec61 and PAT. This density was assigned to EMC, which could be unambiguously fit as a single rigid body based on the characteristic structure of its cytosolic and luminal regions^{26,33–35} (Fig. 3b and Extended Data Fig. 5a,b).

Analysis of the structure reveals how key functional surfaces of EMC are organized at the multipass translocon. On the cytosolic side of

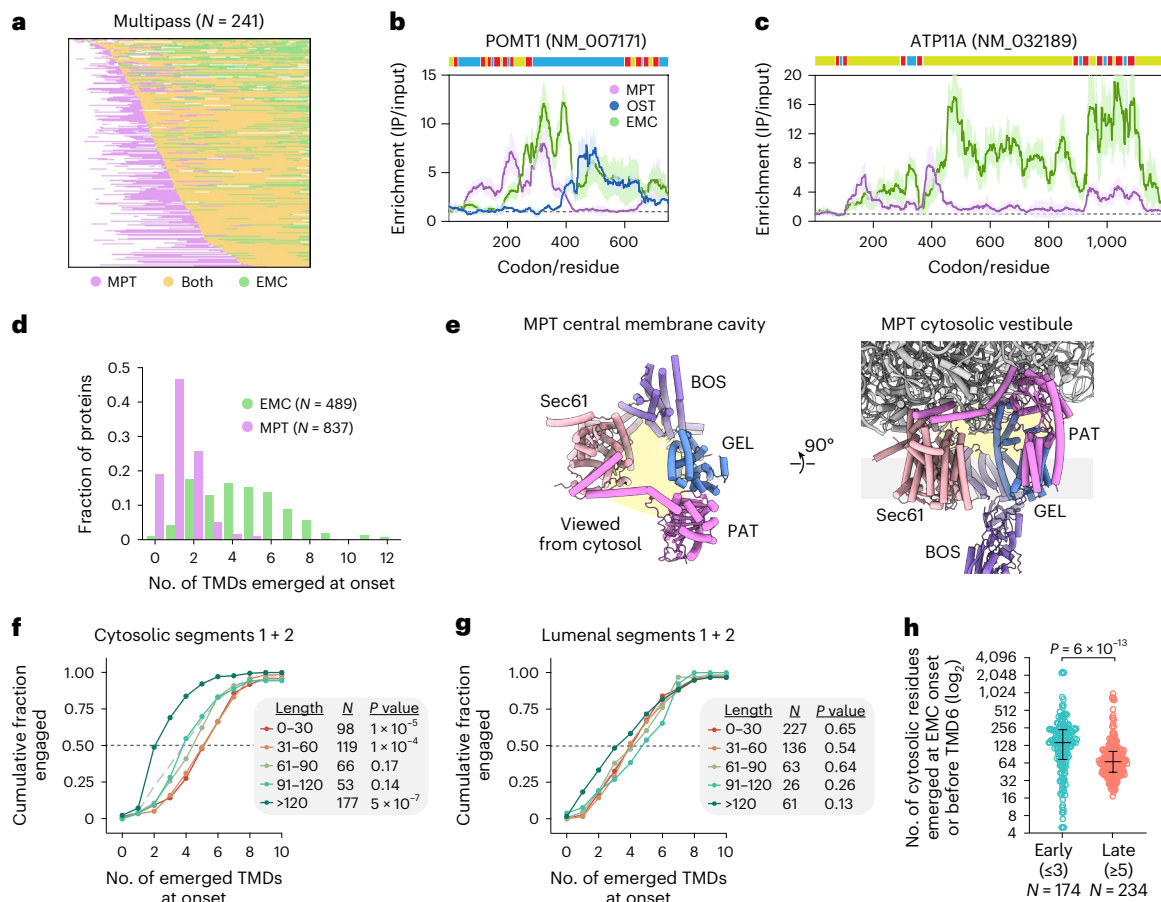


Fig. 2 | Substrate-directed EMC recruitment to the multipass translocon.

a, EMC and MPT peak maps for multipass proteins with gene-level enrichment scores >4 and adjusted P values < 10^{-10} , sorted by the first EMC peak. Each row represents a single transcript normalized by coding region length and colored to indicate positions enriched by EMC (green), MPT (magenta) or both (yellow). **b**, EMC (green), MPT (magenta) and OST-A (blue) interaction profiles for POMT1. Mean enrichment (line) and standard deviation (shaded) are shown. **c**, As in **b**, but for ATP11A. **d**, Distribution of the number of emerged TMDs at the point of MPT (magenta) and EMC (green) onset (assuming 30 residues are buried in the ribosome exit tunnel), for multipass proteins with gene-level enrichment scores >2 and adjusted $P < 10^{-3}$. **e**, Orthogonal views of the multipass translocon (PDB ID 7TUT) highlighting its central membrane cavity (left) and cytosolic vestibule

(right) in yellow. **f**, Cumulative distribution of the number of TMDs emerged at EMC onset for multipass proteins with gene-level enrichment scores >2 and adjusted $P < 10^{-3}$. Proteins were classified by the total number of residues in the first two cytosolic segments. The reference curve (dashed gray) is calculated for all proteins. The number of sequences (N) in each bin, and two-sided Kolmogorov–Smirnov test P values relative to the reference, are indicated. **g**, As in **f**, but for the first two luminal segments. **h**, The log₂ distribution of the number of cytosolic residues emerged at EMC onset (or before TMD6, whichever occurs first) for proteins that recruit EMC early ($\leq$ TMD3) and late ($\geq$ TMD5). The number of sequences (N), median, interquartile range and two-sided Mann–Whitney U test P value are indicated.

the membrane, EMC2, EMC3 and EMC4 pack against portions of the 28S ribosomal RNA (rRNA) and eL31 (Extended Data Fig. 5b). This interface may help stabilize EMC at the ribosome, but it is not sufficient as EMC recruitment is substrate dependent. EMC’s wedge-shaped cytosolic domain allows it to approach closely, creating a large, semi-enclosed cytosolic chamber flanked by the ribosome, EMC and the PAT complex (Fig. 3c). This chamber, which is several-fold larger and more exposed than the MPT cytosolic vestibule (Fig. 2e, right), could shield substrates that can no longer be accommodated within MPT.

On the luminal side of the membrane, EMC1, EMC4, EMC7 and EMC10 are adjacent to (but do not directly contact) the luminal domains of the BOS complex. Weak density was observed for the cytosolic and membrane regions of the TRAP complex, but the luminal regions were not resolved, presumably reflecting their flexibility^{32,36–38}. Thus, the luminal side of the membrane is relatively unobstructed in the presence of EMC, which would allow proteins with long translocated domains to be accommodated.

The membrane-embedded hydrophilic groove formed by the conserved Oxa1 insertase module, EMC3 and EMC6 (possibly gated by

Table 1 | Cryo-ET data collection parameters

	ABCD4-TMD4 programmed HEK293 membranes	
Magnification	×81,000	×81,000
Voltage (kV)	300	300
Total electron exposure (e [−] per Å ²)	100	100
Electron dose per tilt image	2.44	2.44
Defocus range (μm)	−2 to −4	−2 to −4
Pixel size (Å)	1.562	1.562
Energy filter slit width (eV)	10	10
Cryo-focused ion-beam milling angle (°)	N/A	N/A
Start angle (°)	0	0
Tilt range (°)	−60 to +60	−60 to +60
Step size (°)	3	3
Frame number (per tilt)	9	9
Acquisition scheme	Dose-symmetric	Dose-symmetric

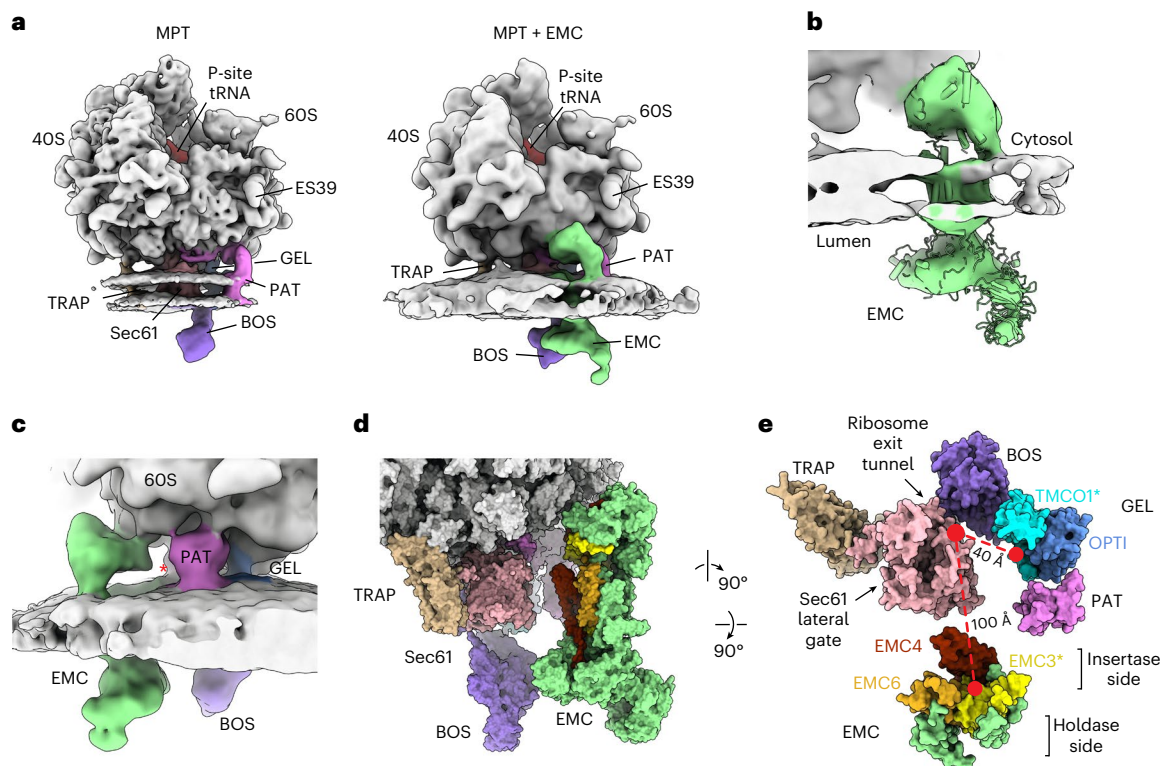


Fig. 3 | Architecture of the EMC-bound multipass translocon. a, Gaussian-filtered, denoised subtomogram averages of the ribosome-bound multipass translocon by itself (left) or bound to EMC (green) (right). Density for the Sec61 (light pink), PAT (magenta), BOS (purple) and GEL (blue) complexes is visible in both maps; density for the TRAP (tan) luminal domains is weak. **b**, Close-up of human EMC (PDB ID 9C7V) docked as a single rigid body into the cryo-ET map. **c**, Close-up of the semi-enclosed cytosolic chamber formed by the large ribosomal subunit (60S), PAT (magenta) and EMC (green). The center of the chamber is marked with an asterisk. The luminal domains of EMC and BOS (purple) are also visible on the opposite side of the membrane. **d**, Surface

representation of a model showing the arrangement of EMC (green) relative to other components of the ribosome-bound multipass translocon. The conserved Oxal insertase subunit EMC3 (yellow), its binding partner EMC6 (goldenrod) and the adjacent EMC4 (maroon) are highlighted. The TRAP luminal region is disordered in the maps, and therefore not shown. **e**, Slice through the membrane of the model in **d**, viewed from the cytosol, showing the expanded lipid-filled cavity formed by EMC docking near the open end of the multipass translocon central cavity. Distances from the ribosome exit tunnel to the conserved Oxal insertase subunits of GEL (TMC01) and EMC (EMC3) are indicated.

EMC4), faces the open end of the multipass translocon central cavity (Fig. 3d,e). This defines an expanded lipid-filled cavity, bounded by EMC, MPT and Sec61. Notably, the GEL and EMC hydrophilic grooves are located on opposite sides, about 40 Å and 100 Å away from the ribosome exit tunnel, respectively (Fig. 3e and Extended Data Fig. 5c,d). Hence, TMDs emerging from the ribosome at the EMC-bound multipass translocon would have at least two potential routes into the membrane, via GEL and EMC. Once inserted, these TMDs could be sequestered in the larger lipid-filled cavity formed at the interface.

The opposite side of EMC houses additional surfaces recently shown to engage an assembly intermediate of the $\text{Ca}_v1.2$ voltage-gated calcium channel^{27,29}. Superimposing the previously determined cryo-EM structure reveals that binding of the 24-TMD pore-forming subunit ($\text{Ca}_v1.2$) and its soluble partner ($\text{Ca}_\beta3$) is fully compatible with EMC assembly at the multipass translocon (Extended Data Fig. 5e). Thus, EMC is positioned at a site where it can interface cotranslationally with nascent membrane proteins emerging from the ribosome and MPT on one side, and posttranslationally with inserted but unassembled substrates on the other (Fig. 3e). During synthesis, substrate TMDs could gain access to the ‘holdase side’ via gaps flanking EMC, which could facilitate cotranslational assembly of multi-subunit membrane protein complexes.

EMC is a cotranslational chaperone at the ER translocon

The structural and profiling analyses indicate that EMC is recruited to the open end of the multipass translocon central cavity when nascent

substrates begin to exceed MPT’s cytosolic or intramembrane capacity. To test this, we used site-specific photocrosslinking to monitor the local interactions of ABCD4 translation intermediates assembled *in vitro* (Fig. 4a). The 2-TMD intermediate containing probes positioned in each of four sequential positions within either TMD1 or TMD2 could be crosslinked to partners of ~17 kDa and ~40 kDa (Fig. 4b, left). Immunoprecipitation showed that the smaller crosslinking partner was TMC01 of the GEL complex and the larger partner was Sec61α (Fig. 4b, right). Thus, following insertion, which occurs independently of the Sec61 lateral gate (Extended Data Fig. 6), the TMD1–TMD2 pair of ABCD4 initially resides within the MPT central cavity between the GEL and Sec61 complexes (Fig. 4c).

These same eight crosslinking probes within the TMD1–TMD2 pair showed a different pattern of crosslinks with a longer, 4-TMD intermediate (Fig. 4d, left). Instead of crosslinks to TMC01 and Sec61α, several new crosslinking partners in the ~20–35 kDa range were observed. These crosslinks could be assigned to EMC3, EMC4, EMC7 and EMC10 by immunoprecipitation (Fig. 4d, right). In this same intermediate, probes located in TMD3, and to a lesser extent TMD4, could be crosslinked to TMC01 of the GEL complex (Fig. 4e), similar to the pattern seen with the TMD1–TMD2 pair in the 2-TMD intermediate. Thus, relative to the 2-TMD intermediate, the TMD1–TMD2 pair in the 4-TMD intermediate has moved away from GEL and close to EMC, whereas the newly inserted TMD3–TMD4 pair now resides near GEL (Fig. 4f, compare to Fig. 4c).

It is notable that with GEL, the interaction is seen from particular TMD residues and not others, following a periodic pattern consistent

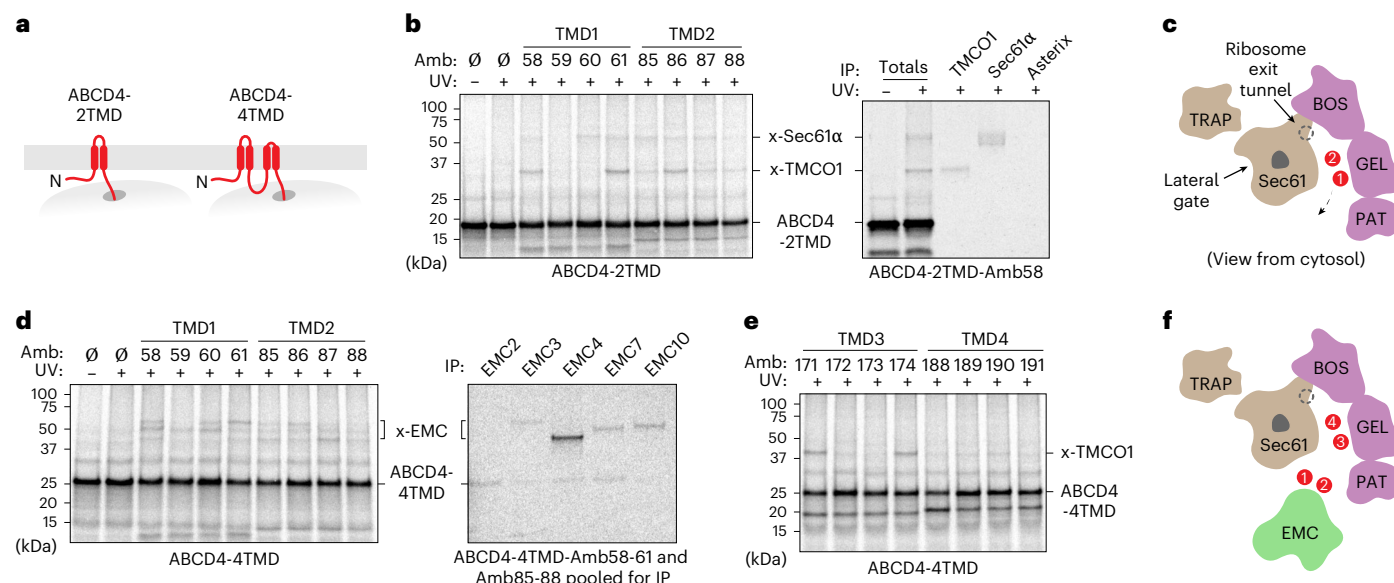


Fig. 4 | EMC engages nascent chain TMDs that exit the multipass translocon central cavity. a, Stalled, Flag-tagged ABCD4 constructs used for site-specific photocrosslinking, truncated 45 residues past the end of TMD2 or TMD4.

b, ³⁵S-methionine labeled ABCD4-2-TMD RNCs containing a UV-activatable benzoyl-phenylalanine at the indicated position (Amb, amber codon) in TMD1 and TMD2 were assembled in the presence of human HEK293 microsomes and analyzed directly by SDS–PAGE and autoradiography (left panel). Crosslinks were validated by immunoprecipitation with the indicated antibody, followed by SDS–PAGE and autoradiography (right panel). **c**, Diagram of ABCD4-2-TMD

showing the initial position of its TMD1–TMD2 pair in the multipass translocon central cavity, based on the data in **b**. **d**, As in **b**, but for ³⁵S-methionine labeled ABCD4-4-TMD RNCs containing a UV-activated benzoyl-phenylalanine at the indicated position in TMD1 and TMD2. **e**, As in **b**, but for ³⁵S-methionine labeled ABCD4-4-TMD RNCs containing a UV-activated benzoyl-phenylalanine at the indicated position in TMD3 and TMD4. **f**, Diagram of ABCD4-4-TMD showing the new location of its TMD1–TMD2 pair on the insertase side of EMC, and its TMD3–TMD4 pair in the multipass translocon central cavity; based on the data in **d** and **e**. All crosslinking experiments were repeated with similar results.

with a helix (Fig. 4b,e). With EMC, different subunits crosslink with different efficiencies from each TMD position (Fig. 4d). These patterns are consistent with direct substrate engagement of both EMC and GEL in favored substrate orientations and positions. We posit that the most hydrophilic faces of each TMD helix are shielded by interactions with each other and by chaperone activities of GEL and EMC. The crosslinking results define the insertase side of EMC as a cotranslational chaperone.

Discussion

We have analyzed cotranslational EMC interactions with the ER-bound ribosome for hundreds of substrates at near-codon resolution, defined the structure of the EMC-bound multipass translocon in native membranes, established the mechanism for EMC recruitment and discovered a cotranslational TMD-chaperoning site on EMC. These data define a set of general principles for the role of EMC during multipass membrane protein synthesis.

First, EMC can be stably recruited to the multipass translocon and, once engaged, remains bound until synthesis is complete. This differs from MPT and OST-A, which can repeatedly engage and disengage from the translocon, particularly during synthesis of large, topologically complex membrane proteins¹². Ribosome-bound EMC does not overlap appreciably with the binding site for MPT and OST-A, nor does it make extensive contact with other components of the translocon (Extended Data Fig. 5g). Because its association is primarily mediated by the nascent chain, EMC is largely insensitive to downstream changes in translocon composition. This explains how already-inserted TMDs remain shielded when MPT disengages during synthesis of very long cytosolic segments, or during translocation of long luminal segments at the secretory translocon.

Second, EMC recruitment coincides with displacement of part or all of the nascent chain from the multipass translocon. This does not strictly require MPT disengagement. Instead, egress is favored when

the number of emerged cytosolic residues or inserted TMDs exceeds the cytosolic or intramembrane capacity of MPT. Presumably, nascent substrates that for some other reason are not well accommodated within the multipass translocon would also egress, resulting in EMC engagement. Thus, the timing of EMC recruitment is substrate-specific, and tuned to the capacity of the multipass translocon to retain the elongating nascent chain within its central cavity.

Third, the substrate ranges of EMC and MPT are only partially overlapping. This is explained by their different modes of recruitment. Whereas MPT preferentially binds to closed Sec61 channels and is therefore recruited during synthesis of nearly all multipass proteins¹², EMC is insensitive to the conformation of Sec61. Instead, EMC is recruited by exposure of already-inserted TMDs. As a result, the subset of multipass proteins whose TMDs can remain sequestered within the multipass translocon central cavity throughout synthesis do not robustly recruit EMC.

Fourth, the conserved Oxa1 superfamily insertase module of EMC can also operate as a TMD chaperone. This is consistent with previous studies on the bacterial Oxa1 superfamily member YidC protein, whose depletion perturbed the folding but not the topology of the multipass sugar transporter LacY^{39,40}. Dual functionality of the hydrophilic groove may be a conserved feature of all Oxa1 superfamily members, including of the GEL complex, as suggested by our crosslinking experiments.

Fifth, the partial redundancy of EMC and GEL as insertases⁴¹ can be explained by their shared ability to engage nascent substrates at the Sec61-bound ribosome. Given its proximity to the ribosome exit tunnel, GEL likely has priority access, with EMC inserting TMD pairs that are skipped by GEL. Our findings may explain why some organisms do not have GEL, whose function could be compensated either by a functionally analogous protein such as SND3 in yeast⁴² or by EMC acting at a membrane-bound ribosome. More work is needed to define the rules for triage between these multiple routes across the membrane.

Last, the EMC holdase surface²⁷ remains fully accessible at the translocon. Following egress, substrate TMDs are first captured on the insertase side of EMC, where the growing polypeptide can be accommodated in a larger, semi-enclosed cytosolic chamber and an expanded lipid-filled cavity at the interface of EMC and MPT. EMC may adjust its position during synthesis to facilitate movement of large multi-domain or oligomeric proteins to the holdase side. This could occur via gaps flanking ribosome-bound EMC, or EMC might transiently release from the ribosome while remaining tethered by the nascent chain. In this way, the multiple activities of EMC as an insertase, TMD chaperone, and holdase could coordinate with MPT to accommodate the widely varying topological and biophysical needs of the membrane proteome.

Online content

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Methods

Antibodies for immunoblotting

Except as noted, antibodies were used at 1:1,000 dilution. Antibodies against human TMC01¹⁰, Sec61 β (1:10,000) and TRAP α (1:5,000)⁴³ were characterized previously. Other antibodies were obtained from the following commercial sources: rabbit anti-CCDC47 (A305-100A), rabbit anti-EMC1 (A305-605A) antibodies from Bethyl Laboratories; goat anti-NOMO1 (PA5-47534) from Thermo Fisher; rabbit anti-mouse IgG H&L (HRP) (ab6728; 1:5,000), rabbit anti-EMC10 (ab180148) and rabbit anti-uL2 (ab169538) antibodies from Abcam; mouse anti-Flag (F1804; 1:2,000), donkey anti-rabbit IgG-peroxidase (SAB3700863; 1:5,000) and rabbit anti-goat IgG-peroxidase (A5420; 1:5,000) antibodies from MilliporeSigma; rabbit anti-EMC4 (27708-1-AP) and rabbit anti-EMC7 (27550-1-AP) antibodies from Proteintech; rabbit anti-Sec61A1 (14868) and rabbit anti-Sec61 β (14648; 1:3,000) antibodies from Cell Signaling Technology; rabbit anti-uL22 antibody from Abgent (AP9892b); mouse anti-OST48 (sc-74408) antibody from Santa Cruz Biotechnology; mouse anti-STT3A (H00003703-M02) antibody from Novus Biologicals; rabbit anti-EMC6 (ARP44679_P050) antibody from Aviva Systems Biology.

Constructs

Coding sequences for EMC7 (NM_020154), EMC10 (NM_175063.6) and ABCD4 (NM_005050.4) were obtained from GenScript (OHu03193D, OHu00818D and OHu19532, respectively); the EMC6 coding sequence (NM_001014764.3) was synthesized as a gBlock by Integrated DNA Technologies. Doxycycline-inducible, N-terminal 3xFlag-tagged EMC6, EMC7 and EMC10 and N-terminal Twin-Strep-tagged EMC10 constructs were generated by PCR amplification followed by Gibson assembly into pcDNA5/FRT/TO (Thermo Fisher, cat. no. V652020) using Phusion polymerase (Thermo Fisher, cat. no. F530L), T5 exonuclease (New England Biolabs, cat. no. M0663S) and Taq Ligase (Fisher Scientific, cat. no. 50-811-617). N-terminal Twin-Strep tagged CCDC47 and OST48 constructs were generated by Gibson cloning to replace N-terminal 3xFlag tags in the corresponding N-terminal Flag-tagged CCDC47 and OST48 constructs in pcDNA5/FRT/TO¹². Affinity tags were added after the signal peptide cleavage site for EMC7, EMC10, CCDC47 and OST48.

Full-length, N-terminally 4xFlag-tagged ABCD4 was Gibson cloned into a pSP64 vector for in vitro translation. Amber stop codon (UAG)-containing ABCD4 constructs for site-specific photocrosslinking were generated by site-directed mutagenesis with PfuUltra HF DNA polymerase (Agilent Technologies, cat. no. 600382). All constructs were confirmed by DNA sequencing.

Cell culture and cell line construction

All experiments used Flp-In T-REx 293 cells (Thermo Fisher, cat. no. R78007), grown at 37 °C under 5% CO₂ in DMEM (Corning, cat. no. MT10013CV) containing 10% fetal bovine serum (GeminiBio, 900-108), 100 U ml⁻¹ penicillin and 100 μ g ml⁻¹ streptomycin (GeminiBio, cat. no. 400-109). Cells were tested for mycoplasma at approximately 6-month intervals using the Universal Mycoplasma Detection Kit (American Type Culture Collection, cat. no. 30-1012K) and were negative throughout.

Stable, doxycycline-inducible Flp-In T-REx 293 cell lines were generated in the corresponding wild-type (SS_{ER}-3xFlag-EMC7, SS_{ER}-3xFlag-EMC10, SS_{ER}-Twin-Strep-EMC10, SS_{ER}-Twin-Strep-CCDC47 and SS_{ER}-Twin-Strep-OST48) or knockout (3xFlag-EMC6) background¹⁶ according to the manufacturer's instructions (Thermo Fisher). Briefly, 100 ng of the desired plasmid were cotransfected with 900 ng of pOG44 (1:9 ratio) in a six-well plate using TransIT (Mirus Bio, cat. no. MIR2700). After 24 h, cells were selected for Flp-mediated recombination with 100 μ g ml⁻¹ hygromycin B (Gibco, cat. no. 10687010) for 3–4 weeks to obtain the stably integrated cell lines.

Sample preparation for selective ribosome profiling

Flag-tagged EMC6, EMC7 and EMC10 samples were prepared as described previously in ref. 12. Briefly, cells were seeded in ten 15-cm

dishes, induced with 0.1–1.0 ng ml⁻¹ doxycycline (MP Biomedicals, cat. no. 195044) and collected after 72 h (~80% confluency) with ice-cold PBS (Corning, cat. no. 21040CV) containing 0.1 mg ml⁻¹ cycloheximide (CHX) (Sigma, cat. no. 01810). Cells were recovered by centrifugation at 2,000g for 5 min at 4 °C, washed once with ice-cold PBS containing 0.1 mg ml⁻¹ CHX, collected by centrifugation, snap frozen and stored at –80 °C.

Cells were lysed in 9 ml of hypotonic homogenization buffer (10 mM HEPES-KOH pH 7.5, 10 mM KOAc, 1 mM MgCl₂ and 0.5 $\times$ protease inhibitor cocktail (Roche, cat. no. 11836170001)) containing 0.1 mg ml⁻¹ CHX for 20 min on ice. Cells were then homogenized by 25 strokes (up and down) in a chilled dounce tissue grinder. Sucrose was added to a final concentration of 250 mM and mixed gently. Nuclei and cell debris were removed by centrifugation at 1,500g for 15 min at 4 °C and the supernatant collected. The supernatant fractions were centrifuged again at 1,500g for 10 min at 4 °C. The supernatant was collected and centrifuged at 16,500g for 15 min at 4 °C. The resulting membrane pellet was washed once with 2 ml of insertion buffer (50 mM HEPES-KOH pH 7.5, 250 mM sucrose, 250 mM KOAc, 10 mM MgCl₂) containing 0.1 mg ml⁻¹ CHX, and resuspended in the same buffer to an absorbance at 260 nm (A₂₆₀) of 50.

Microsomes (A₂₆₀ of 50) were treated for 30 min at 37 °C with 5 μ l (10,000 U) micrococcal nuclease (NEB, cat. no. M0247S), 1.25 mM CaCl₂ and 0.5 mM PMSF. After micrococcal nuclease treatment, 3 μ l (3 U) RNase-Free DNase (Promega, cat. no. M6101) was added, incubated for 5 min at room temperature and quenched with 2.5 mM EGTA. Microsomes were pelleted at 13,500g for 10 min at 4 °C. The supernatant was discarded, and the membrane pellet was solubilized with insertion buffer supplemented with 2.5% digitonin (Millipore, cat. no. 300410), 0.1 mg ml⁻¹ CHX and 1 $\times$ protease inhibitor cocktail for 45 min on ice. The digitonin-solubilized material was diluted twice with 150 mM KOAc insertion buffer containing 0.1 mg ml⁻¹ CHX and cleared by centrifugation at 13,500g for 15 min at 4 °C. The cleared supernatant (which served as the 'Input' sample for ribosome profiling) (A₂₆₀ of ~5) was immunoprecipitated in batch using 50 μ l of M2 Flag affinity beads (Sigma, cat. no. A2220) and end-over-end mixing overnight at 4 °C. Flow-through was removed and beads were washed 4 times with 14 column volumes of insertion buffer containing 0.4% digitonin and 0.1 mg ml⁻¹ CHX. Bound material was eluted twice, for 30 min on ice, with 1.5 column volumes of 200 mM KOAc insertion buffer supplemented with 0.5 mg ml⁻¹ Flag peptide (ApexBio, cat. no. A6001), 0.4% digitonin and 0.1 mg ml⁻¹ CHX. The eluate from 3xFlag-EMC7 and 3xFlag-EMC10 (which served as the 'IP' (immunoprecipitation) sample for ribosome profiling) was collected using a pre-equilibrated spin filter column (Thermo Fisher, cat. no. 69725), snap frozen and stored at –80 °C. The Flag peptide eluate from 3xFlag-EMC6 was additionally concentrated by pelleting the ribosomes through a 300- μ l sucrose cushion (50 mM HEPES pH 7.4, 10 mM MgCl₂, 150 mM KCl, 500 mM sucrose and 0.4% digitonin) at 355,000g for 1 h at 4 °C in a TLA120.1 rotor. The ribosome pellet was resuspended in 80 μ l of 200 mM KOAc insertion buffer supplemented with 0.4% digitonin and 0.1 mg ml⁻¹ CHX, snap frozen and stored at –80 °C.

Rfoot-seq library preparation

Libraries for the Input and immunoprecipitation EMC samples were prepared by Rfoot-seq essentially as described in refs. 12,30,31,38. Briefly, RNA concentration was measured using the Qubit RNA high-sensitivity assay (Invitrogen, cat. no. Q32852), and RNase footprints were prepared as follows. For Input samples, 800–900 ng RNA were digested with 90 U of RNase I (LGC Biosearch Technologies, N6901K) in a 95- μ l reaction. For immunoprecipitation samples, 300–500 ng RNA were digested with 40 U of RNase I in a 90- μ l reaction. The reaction volume was adjusted with lysis buffer (20 mM Tris pH 7.4, 150 mM NaCl, 5 mM MgCl₂, 1 mM dithiothreitol (DTT), 0.1 mg ml⁻¹ CHX, 1% v/v Triton X-100). The mixture was incubated at room temperature (25 °C) for 1.5 h with gentle agitation on a nutating mixer, quenched with 400 μ l of TRIzol (Ambion, cat. no. 15596026), and vortexed thoroughly. After adding

100 μ l of chloroform (Sigma-Aldrich, cat. no. C2432), RNA in the aqueous layer was separated by centrifugation at 12,000g for 15 min at 4 °C and precipitated overnight by adding 0.1 volume of 3 M sodium acetate (Invitrogen, cat. no. AM9740), 20 μ g of GlycoBlue (Invitrogen, cat. no. AM9515) and 1.2 volumes of isopropanol.

Purified RNase footprints used for DNA library preparation were resuspended in 5 μ l of T4 polynucleotide kinase (PNK) reaction mixture (1 $\times$ PNK buffer, 10 U of PNK (NEB, cat. no. M0201L) and 10 U of SUPERaseIn (Invitrogen, cat. no. AM2694)) and end-repaired at 37 °C for 1 h. Next, 5 μ l of polyadenylation mix containing 10 U of *Escherichia coli* poly (A) polymerase (NEB, cat. no. M0276L), 2 μ l of 5 $\times$ first-strand buffer (250 mM Tris-HCl pH 8.3, 375 mM KCl, 15 mM MgCl₂), 0.25 mM ATP and 10 U SUPERaseIn was added, and the reaction was incubated at 37 °C for 2 h. The resulting polyadenylated RNAs were reverse transcribed by SMART-RT, by adding 2.5 μ l of reverse transcription primer (10 μ M, 5'-ACGTGTGCTCTCCGATCTNNNNNNNNNNNNNNNTTTTTTTTTTTTTTTT-3', where 'N' is any base and 'V' is 'A', 'C' or 'G'), annealing by heating to 72 °C for 5 min and cooling on ice, and then supplementing with 17.6 μ l of SMART-RT mixture (2.5 μ l of dNTP (10 mM), 400 U of Superscript II reverse transcriptase (Invitrogen, cat. no. 18064014), 10 U of SUPERaseIn, 4 μ l of Superscript II first-strand buffer (5 $\times$), 1.5 μ l of DTT (100 mM), 6 μ l of Betaine (5 M), 0.12 μ l of MgCl₂ (1 M), 0.5 μ l of template-switching oligos (100 μ M 5'-CTCTTCCCTACACGAGCTCTTCCGATCTNNNNrGrG+G-3', where 'N' is any base and '+G' is a locked nucleic acid guanylate) and 0.5 μ l of nuclease-free water). Reverse transcription was carried out at 42 °C for 1 h, followed by 10 cycles of heating at 50 °C for 2 min and incubating at 42 °C for 2 min. Reverse transcriptase was inactivated by heating at 70 °C for 10 min. RNA footprints hybridized to DNA were removed by incubating with 5 U of RNase H (NEB, cat. no. M0297L) for 15 min at 37 °C and the reaction was inactivated by heating at 65 °C for 15 min.

The resulting complementary DNA (30 μ l) was amplified by a first round of PCR in a 100- μ l reaction containing 2 $\times$ Ultra II Q5 Master Mix (NEB, cat. no. M0544L), 0.5 μ M PCR1 forward primer (5'-GATCTACACTCTTCCCTACACGAGC-3') and 0.5 μ M PCR1 reverse primer (5'-GTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT-3') for 7 cycles. After purification (Zymo Research, cat. no. D4014), the sample was subjected to a second round of PCR in a 50- μ l reaction using 0.5 μ M PCR2 forward primer (5'-AATGATACGGCGACCACCGAGATCTACACTCTTCCCTACAC-3') and 0.5 μ M PCR2 reverse primer (5'-CAAGCAGAAGACGGCATACGAGATNNNNNNGTGACTGGAGTTCAGACGTGTG-3') containing the N₆ unique barcoded library index, for 4 cycles. After purification, the DNA library was visualized and separated in a 4% agarose gel. DNA with a 15–35 base-pair insert was excised and recovered using a DNA gel recovery kit (Zymo Research, cat. no. D4008). Equal amounts of barcoded libraries were pooled and then sequenced with Illumina NovaSeq.

Rfoot-seq data processing and analysis

Ribosome profiling data were processed and analyzed essentially as described previously in ref. 12. Briefly, reads were processed by trimming adapters with Cutadapt v.4.1, removing rRNA reads with Bowtie v.2.2.6⁴⁴ and aligning remaining reads to the human hg38 genome and RefSeq-defined transcriptome with TopHat v.2.1.0⁴⁵. Only uniquely mapped reads were used for downstream analyses. High-quality reads showing strong 3-nucleotide periodicity were selected and adjusted to the ribosomal A-site using RibORF^{46–48}.

The hg38 genome sequence and RefSeq transcript annotations were retrieved from the UCSC Genome Browser. Where multiple transcript isoforms existed for a gene, the longest coding isoform was selected as the representative isoform. Protein sequences were processed using DeepTMHMM⁴⁹ to predict TMDs and membrane topology. Mitochondrial and tail-anchored proteins were excluded, and proteins lacking a signal peptide or TMD (cytonuclear) were used as the background gene set.

Transcript abundance was quantified as transcripts per million using HTSeq-count v.2.0.3⁵⁰. Transcript enrichment was calculated as the log₂ ratio of transcripts per million values between the immunoprecipitation and Input conditions, normalized by median-centering to the cytonuclear protein background. Statistical significance was determined via chi-square tests comparing gene-level counts against the background mean, with *P* values adjusted using the Benjamini–Hochberg correction. Positional enrichment scores ($E_{i,j}$) were calculated at codon resolution as previously described in ref. 12. Briefly, A-site-adjusted reads were extended by 50 nt to smooth local occupancy, and scores were normalized to background cytonuclear genes. Interacting regions (peaks) were defined as continuous stretches of at least 20 codons satisfying $E_{i,j} > 4$ and an adjusted $P < 10^{-4}$. The analyses shown in Fig. 2 were robust to lower ($E_{i,j} > 3$ and adjusted $P < 10^{-3}$) and higher ($E_{i,j} > 5$ and adjusted $P < 10^{-5}$) stringency criteria.

Sucrose gradient

Micrococcal nuclease-treated microsomes were solubilized with a low sucrose (150 mM) insertion buffer supplemented with 0.1 mg ml⁻¹ CHX, 2.5% digitonin and 1 $\times$ protease inhibitor cocktail for 30 min on ice. The digitonin-solubilized material was diluted twice with 150 mM KOAc, 150 mM sucrose insertion buffer containing 0.1 mg ml⁻¹ CHX and cleared by centrifugation at 16,500g for 15 min at 4 °C. The cleared supernatant was laid over 1 ml of sucrose (1 M) cushion insertion buffer containing 150 mM KOAc and 0.4% digitonin, and centrifuged at 440,000g for 1 h at 4 °C in a TLA100.3. Ribosome pellets were resuspended in 400 μ l of low sucrose (100 mM) insertion buffer containing 200 mM KOAc, 0.1 mg ml⁻¹ CHX and 0.4% digitonin, and overlaid on a sucrose gradient (10–35% sucrose (wt/vol), 50 mM HEPES pH 7.4, 5 mM MgCl₂, 150 mM KOAc and 0.1% digitonin) and centrifuged at 150,000g in an SW28.1 rotor for 8 h at 4 °C. In total 17 $\times$ 1-ml fractions were collected, TCA precipitated, resuspended in sample loading buffer and fractions 3 to 16 analyzed by SDS–PAGE and immunoblotting.

Interaction analysis of stalled RNCs in vitro

Microsomes used for in vitro translation were prepared essentially as described above for profiling, but without CHX. Briefly, cells were lysed in 9 ml of hypotonic homogenization buffer (10 mM HEPES-KOH pH 7.5, 10 mM KOAc, 1 mM MgCl₂ and 0.5 $\times$ protease inhibitor cocktail (Roche, cat. no. 11836170001)) for 20 min on ice. Cells were then homogenized by 25 strokes (up and down) in a chilled dounce tissue grinder. Sucrose was added to a final concentration of 250 mM and mixed gently. Nuclei and cell debris were removed by centrifugation at 1,000g for 10 min at 4 °C and the supernatant collected. The pellet was resuspended in 2 ml of insertion buffer (50 mM HEPES-KOH pH 7.5, 250 mM sucrose, 250 mM KOAc, 10 mM MgCl₂) and centrifuged again at 1,000g for 10 min at 4 °C, and the supernatant collected. The pooled supernatant was centrifuged again at 1,000g for 10 min at 4 °C. The supernatant was collected and centrifuged at 13,500g for 10 min at 4 °C. The supernatant was discarded, membrane pellet was washed once with 2 ml of insertion buffer and the resulting membrane pellet was resuspended in the same buffer to a final A₂₆₀ of 25.

Microsomes were treated for 15 min at 37 °C with 5,000 U of micrococcal nuclease (NEB, cat. no. M0247S), 1.25 mM CaCl₂ and 0.5 mM PMSF and 2.5 U of RNase-Free DNase (Promega, cat. no. M6101). Micrococcal nuclease treatment was then quenched with 2.5 mM EGTA and kept on ice for 5 min before centrifugation at 13,500g for 10 min at 4 °C. Supernatant was removed completely and resuspended with 1 ml of insertion buffer containing 50 U of SUPERaseIn (Invitrogen, cat. no. AM2694) and 0.125 mM EGTA and incubated on ice for 5 min. Microsomes were centrifuged again at 13,500g for 10 min at 4 °C. Supernatant was completely removed and resuspended again with 1 ml of insertion buffer containing 50 U of SUPERaseIn and incubated on ice for 5 min. Microsomes were centrifuged again at 13,500g for 10 min at 4 °C. The preparation was finally adjusted with insertion buffer to a

final concentration of A_{260} of 50, and 120- μ l aliquots were flash frozen in liquid nitrogen and stored at -80°C for further analysis.

DNA templates for the synthesis of N-terminally Flag-tagged substrates truncated 45 residues past the end of the desired TMD, including a terminal Met–Leu–Lys–Val sequence and no stop codon, were amplified by PCR. In vitro transcription and translation was performed essentially as described in ref. 12. Briefly, 100 μ l of in vitro transcription mix containing 500 ng of purified PCR template was incubated for 1 h at 40°C . Next, 500 μ l of translation reactions containing 100 μ l of the unpurified transcription reaction and 50 μ l of treated HEK293-derived microsomes (A_{260} of 50) were incubated for 50 min at 32°C . The translation reactions were stopped by diluting with 600 μ l of IVT stop buffer (50 mM HEPES pH 7.5, 200 mM NaCl and 10 mM MgCl_2) and microsomes were pelleted at 13,500g for 10 min. Microsomes were resuspended with a 500 μ l of IVT stop buffer containing 200 mM sucrose, 5,000 U of micrococcal nuclease (NEB), 1.25 mM CaCl_2 and 0.5 mM PMSF, and incubated at room temperature for 15 min. The reaction was diluted with 500 μ l of IVT stop buffer, quenched with 2.5 mM EGTA and centrifuged at 13,500g for 10 min. The resulting membrane pellet was solubilized with 200 μ l of insertion buffer supplemented with 2.5% digitonin and 1 $\times$ protease inhibitor cocktail (Roche, cat. no. 11836170001) for 30 min on ice, then diluted twofold with 150 mM KOAc insertion buffer and cleared by centrifugation at 13,500g for 15 min.

The cleared supernatant (A_{260} of 1.0) was incubated overnight at 4°C with 20 μ l of anti-Flag M2 affinity resin (Sigma, cat. no. A2220). Flow-through was removed and beads were washed 3 times with 20 column volumes of insertion buffer containing 0.4% digitonin and 200 mM KOAc. Bound material was eluted twice with 2 column volumes of 200 mM KOAc insertion buffer supplemented with 0.5 mg ml^{-1} Flag peptide and 0.4% digitonin, by incubating for 30 min each on ice. The eluate was collected using a pre-equilibrated spin filter column (Thermo Fisher, cat. no. 69725) and normalized by A_{260} . Where noted, the eluate was additionally treated with 50 $\mu\text{g ml}^{-1}$ RNaseA (Thermo Fisher, cat. no. EN0531), and 10 mM EDTA for 20 min at room temperature to digest the peptidyl-tRNA ester bond before analysis by SDS–PAGE and immunoblotting.

In the case of tandem Strep pull-down from the initial Flag immunoprecipitation, the normalized Flag peptide elution was diluted twice with insertion buffer containing 0.4% digitonin and 200 mM KOAc. The eluate was gently rotated for 2 h with 20 μ l of Strep-TactinXT beads (IBA, cat. no. 2-5030-002). Flow-through was removed and beads were washed three times with 15 column volumes of insertion buffer containing 0.4% digitonin and 200 mM KOAc. Beads were heated with 60 μ l of 2 $\times$ LDS sample buffer containing 2% β -mercaptoethanol, and analyzed by SDS–PAGE and immunoblotting.

Site-specific photocrosslinking

Preparation of ER microsomes, in vitro transcription and translation analysis for site-specific photocrosslinking was performed as described previously¹⁹. *p*-benzoyl-L-phenylalanine (Bpa) was incorporated at specific positions encoded by amber stop codon (UAG) by adding 5 μM suppressor *Bacillus stearothermophilus* transfer RNA (tRNA), 0.25 μM Bpa tRNA synthetase purified from *E. coli* and 100 μM Bpa (BACHEM, cat. no. 4017646) to the translation reaction. A standard translation reaction was performed at 32°C for 30 min.

Photocrosslinking was typically performed after translation reactions. After translation, membranes were pelleted at maximum speed in a tabletop centrifuge (Eppendorf, cat. no. 5424R) at 4°C . Pelleted membranes were washed with 1 $\times$ ribosome–nascent chain complex (RNC) buffer (50 mM HEPES, pH 7.4, 100 mM KOAc, 5 mM $\text{Mg}(\text{OAc})_2$) and resuspended in 0.5 $\times$ RNC buffer. Resuspended membranes were transferred to precooled 384-well plates (Thermo Fisher, cat. no. 4309849) kept on a cold ice block, and subjected to ultraviolet (UV) irradiation using a UVP Blak-Ray B-100AP High-Intensity Lamp with the bulb positioned 10 cm above the samples. After UV crosslinking, samples were

returned to ice, and a final concentration of 50 $\mu\text{g ml}^{-1}$ RNaseA (Thermo Fisher, cat. no. EN0531) and 10 mM EDTA (Sigma, cat. no. E4884) were added to digest peptidyl-tRNA ester bonds. Approximately 1.5 μ l of crosslinked samples (with matched samples not exposed to UV where indicated) were analyzed. Where applicable, approximately 20 μ l of crosslinked samples were subjected to immunoprecipitation with protein A beads (Repligen, cat. no. CA-HF-0100) and the following antibodies: anti-Sec61A1 (5 μ l, Cell Signaling Technology, cat. no. 14868), anti-Asterix (2 μ l, Thermo Fisher, cat. no. PA5-66788), anti-TMCO1 (2 μ l, Abcam, cat. no. ab220729), anti-EMC2 (2 μ l, Proteintech, cat. no. 25443-1-AP), anti-EMC3 (2 μ l, Thermo Fisher, cat. no. 711771), anti-EMC4 (2 μ l, Bethyl, cat. no. A305-752A-M), anti-EMC7 (2 μ l, Bethyl, cat. no. A305-678A-M) and anti-EMC10 (2 μ l, Abcam, cat. no. ab180148).

Protease protection assays

Ten-microliter translation reactions were used for analysis of inserted ABCD4 via proteinase K (Fisher Scientific, cat. no. BP1700-100). As indicated, apratoxin A (a gift from V. Paavilainen and K. McPhail) was added to the translation reaction at a final concentration of 2 μM . Following translation, membranes were collected by centrifugation at maximum speed in a tabletop centrifuge at 4°C and resuspended in 0.5 $\times$ RNC buffer. A final concentration of approximately 0.5 mg ml^{-1} proteinase K was added and digested for 50 min on ice. This reaction was quenched by adding 5 mM PMSF (Sigma, cat. no. 78830) and transferring to boiling 1% SDS, 100 mM Tris-HCl, pH 8.0 for denaturation. Equal amounts of samples before and after proteinase K digestion were analyzed.

Cryo-ET sample preparation and data collection

In vitro translation (150 μ l of total reaction volume) was performed as described above (section ‘Interaction analysis of stalled RNCs in vitro’). Translation reactions were supplemented with 30 μ l of the transcription reaction for the stalled Flag-ABCD4-4-TMD truncation (TMD4 + 45 residues, including the terminal MLKV) and 15 μ l of micrococcal nuclease-treated HEK293 microsomes to a final A_{260} of 4–6. After 50 min, reactions were stopped on ice with 900 μ l of IVT stop buffer (50 mM HEPES pH 7.5, 200 mM NaCl, 5 mM MgCl_2). Microsomes were pelleted (Eppendorf Centrifuge, cat. no. 5417R) at 10,000g for 10 min at 4°C , washed once with 500 μ l of IVT stop buffer and resuspended to a final concentration of 9 mg ml^{-1} in resuspension buffer (10 mM HEPES pH 7.5, 1 mM MgCl_2 , 1 mM glutathione (Thermo Fisher, cat. no. A1801414)).

Programmed microsomes were diluted to 0.15–0.35 mg ml^{-1} in a 1:1 solution of resuspension buffer and 10-nm BSA-treated gold fiducial beads (Aurion Gold Nanoparticles, Electron Microscopy Sciences, cat. no. 25487). For grid freezing, 3.5 μ l of the premixed microsomes and gold fiducials were applied to glow discharged Quantifoil R 3.5/1 200 Mesh Copper or Gold grids (Electron Microscopy Sciences, cat. nos. Q250CR-35 or Q250AR-35) and incubated on the grids for 20 s. After incubation, the grids were back-blotted for 15 s and plunge-frozen in liquid ethane using a Leica EM GP2 (Leica Microsystems). The Leica EM GP2 was set to 22°C and 99% relative humidity. Plunge-frozen grids were then clipped and stored in cryo-grid boxes under liquid nitrogen.

A total of 497 tilt series were collected on a 300 kV Titan Krios G4i cryo-transmission electron microscope (Thermo Fisher) equipped with a Falcon 4i direct electron detector (Thermo Fisher) and a SelectrisX energy filter (10 eV slit width) with SerialEM⁵¹. Tilt series were collected with a dose-symmetric tilt scheme⁵² from 0° in 3° increments, groups of two, from -60° to $+60^{\circ}$ with a total dose of $-100\text{ e}^{-}/\text{\AA}^2$. The unbinned pixel size was 1.562 $\text{\AA}$ and the nominal defocus was between $-2\text{ }\mu\text{m}$ and to $-4\text{ }\mu\text{m}$ in 0.25- μm steps. The raw frames were saved in the Electron Event Representation (EER) format. Full data collection parameters are given in Table 1.

Tomogram reconstruction

The cryo-ET data processing workflow is outlined in Extended Data Fig. 4. In total, 144 EER frames were collected per tilt image. The raw EER data

were fractionated into 9 groups (16 frames per fraction) and subjected to motion correction and contrast transfer function (CTF) estimation in Warp⁵³ (v.2.0.0). The resulting motion-corrected frames were then compiled into tilt series image stacks and aligned using fiducial alignment in IMOD⁵⁴ (v.4.12.52). CTF estimation for the entire tilt series was performed in Warp, and full tomograms were reconstructed by weighted back projection at a pixel size of 20 Å.

Particle picking

Particle coordinates were determined using pytom-match-pick⁵⁵ (v.0.11.0) template matching against a reconstruction of a mammalian 80S ribosome (EMD-2938), low-pass filtered to 40 Å and downsampled to match the tomogram pixel size (20 Å) using pyTOM. The ribosome positions were used to extract three-dimensional (3D) volumes and corresponding CTF volumes at a pixel size of 4.2 Å (2.7× binned) in Warp.

Subtomogram analysis

The 45,320 extracted subtomograms were refined in RELION⁵⁶ (v.5.0.1) using a spherical mask with a diameter of 450 Å against a 3D reference map of an 80S ribosome reconstructed from the full dataset. The aligned particles were subject to 3D classification with a spherical mask with a diameter of 700 Å against the map from the initial 3D refinement job low-pass filtered to 40 Å ($T = 4$, classes 5) to remove junk particles and false positives. The remaining 44,607 particles were refined in RELION and used for subsequent focused classification to identify different ribosome–translocon complexes.

Initial classification of ER ribosome populations

The refined particles were subject to 3D classification in RELION using a soft mask focused on the secretory translocon against a 3D reference map from the initial refinement job that was low-pass filtered to 20 Å ($T = 4$, classes 9). Particles were sorted into secretory translocon-bound ribosomes, MPT-bound ribosomes, soluble ribosomes and a combined class of ribosomes with ambiguous densities. Ribosomes with ambiguous densities were subject to two additional rounds of classification and sorted to the respective classes described above until no further separation could be achieved. Subtomograms of the secretory translocon class and the soluble class were refined in RELION and then subject to iterative refinement in M (v.2.0.0)⁵⁷ using the reconstructions of the two half maps as a reference and a tight, soft mask of the whole particle at a pixel size of 2.1 Å (1.3× binned). Particles were subjected to three iterations of refining the image warp grid, particle poses, stage angles, volume warp grid, defocus and pixel size.

Classification of MPT-bound ribosomes

Subtomograms of the MPT class (box size 200 voxels) were recentered by ~13 nm from the center of the ribosome toward Sec61, refined in RELION and subjected to further classification using a soft mask focused on Sec61, Nicalin, TMEM147 and CCDC47 ($T = 4$, classes 5). The low-pass filtered map from the refinement of the MPT particles was used as reference. Particles were sorted into MPT-bound ribosomes and core translocon-bound ribosomes (that is, Sec61 and TRAP only). Subtomograms of the core translocon class were refined in RELION and then subject to iterative refinement in M using the reconstructions of the two half maps as a reference and a tight soft mask of the whole particle at a pixel size of 2.1 Å (1.3× binned). Particles were subject to 2–3 iterations of refining the image warp grid, particle poses, stage angles, volume warp grid, defocus and pixel size.

The remaining subtomograms of the MPT class were further refined in RELION and then used for focused classification using a soft mask around CCDC47 and an additional weak density corresponding in size, shape and location to EMC ($T = 4$, classes 7). A low-pass filtered map from the refinement of the MPT class was used as a reference. Particles were sorted into an MPT class and an MPT + EMC class. Each class was refined in RELION and then subject to iterative refinement in

M using the reconstructions of the two half maps as a reference and a tight soft mask of the whole particle at a pixel size of 2.1 Å (1.3× binned). Particles were subject to 2–3 iterations of refining the image warp grid, particle poses, stage angles, volume warp grid, defocus and pixel size. Final maps for the MPT and MPT + EMC classes were generated in M and used for visualization as indicated in the figure legends.

The EMC-bound MPT model shown in Fig. 3 and Extended Data Fig. 5 was generated by rigid-body docking of the previously determined structure of human EMC (Protein Data Bank (PDB) ID 9C7V) and a trimmed AlphaFold3 model of human TRAP into the low-resolution cryo-ET map. The remaining human translocon components (Sec61, GEL, PAT and BOS) were placed by superimposing trimmed AlphaFold3 models onto the cryo-EM structure of the canine MPT (PDB ID 7TUT) docked into the cryo-ET map. The composite model was used for visualization purposes only and was not deposited.

Statistics and reproducibility

Selective ribosome profiling was performed in two biological replicates for 3xFlag-EMC6 and 3xFlag-EMC7, and three biological replicates for 3xFlag-EMC10. Transcript enrichment was reproducible between biological replicates and highly correlated between the three tagged subunits; the seven profiling datasets were therefore combined into a single EMC sample ($n = 7$). Cryo-ET reconstructions were obtained by combining five datasets from four independently prepared samples. All in vitro analyses are representative of at least two independent experiments, performed in part or in full, with similar results.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Sequencing data are available at the National Center for Biotechnology Information Gene Expression Omnibus repository with accession number GSE319640. Cryo-ET maps are available at the Electron Microscopy Data Bank with accession codes EMD-75661 (ribosome-bound MPT) and EMD-75658 (ribosome-bound MPT with EMC). All other data are available in the main text or the supplementary materials. Source data are provided with this paper.

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Author contributions

J.T. and A.S. prepared samples for selective ribosome profiling and performed the in vitro interaction analyses. Q.L. built libraries for ribosome profiling. Y.W., with guidance from Z.J. and R.J.K., processed

and analyzed the ribosome profiling data. M.G.R., with additional guidance from J.F., determined the cryo-ET structures. H.W. performed the site-specific photocrosslinking experiments. A.S., Z.J. and R.J.K. conceived of the project. R.S.H., Z.J. and R.J.K. supervised the project and provided funding. R.J.K. wrote an initial draft of the paper, which was revised with input from all authors.

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Competing interests

R.S.H. is a consultant for and equity holder in Gate Biosciences. The other authors declare no competing interests.

Additional information

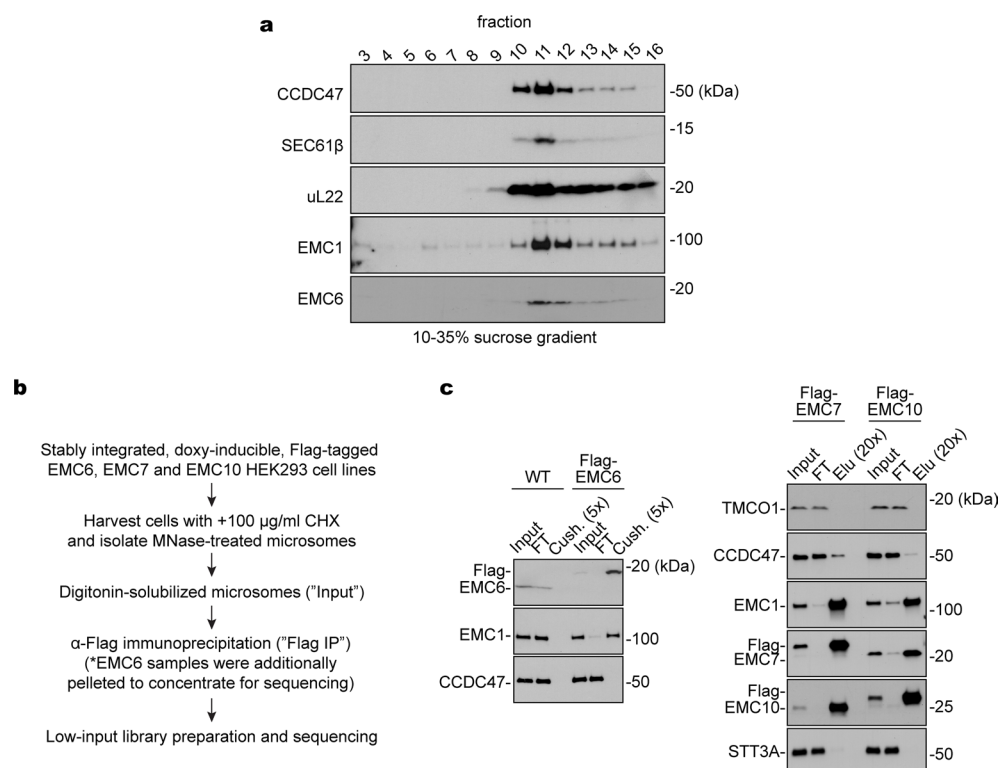
Extended data is available for this paper at <https://doi.org/10.1038/s41594-026-01889-2>.

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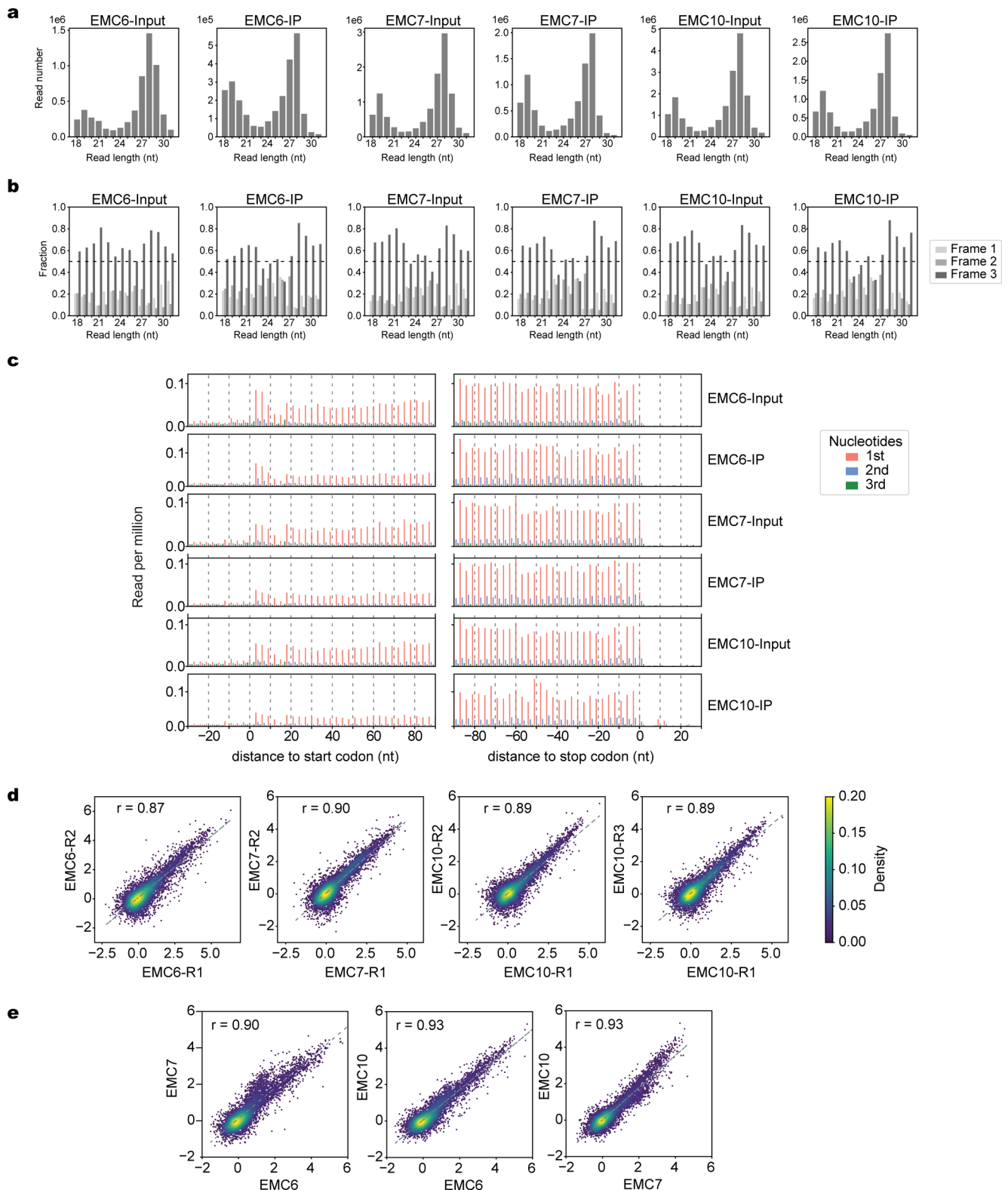
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Extended Data Fig. 1 | Sample preparation for selective ribosome profiling.

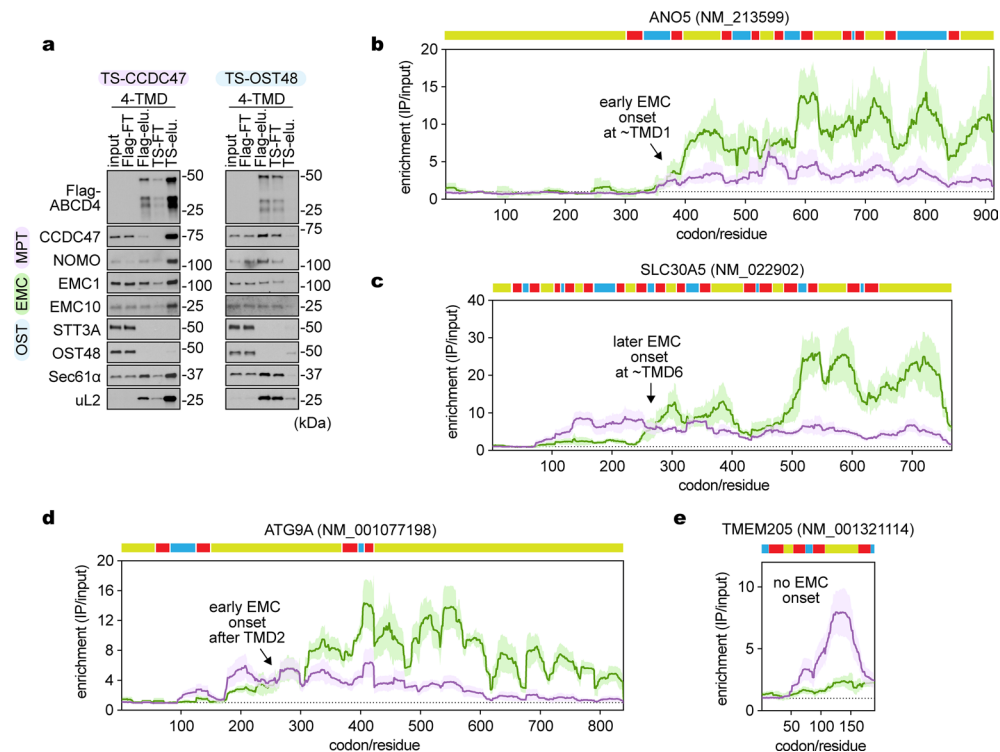
(a) Micrococcal nuclease (MNase)-treated, digitonin-solubilized HEK293 membranes were fractionated by sucrose cushion, separated on a 10-35% sucrose gradient, and analyzed by immunoblotting. (b) Overview of the sample preparation strategy for selective ribosome profiling. (c) Immunoblotting of MNase-treated, digitonin-solubilized microsomes ('input') and the corresponding flow-through (FT), cushion (Cush) or elution (Elu) fractions following affinity purification via the indicated Flag-tagged EMC subunit.

Doxy-induced Flag-EMC7 was expressed at near endogenous levels (compare with the endogenous EMC7 level in Flag-EMC10 cells); Flag-EMC10 was expressed at slightly elevated levels (compare with the endogenous EMC10 level in Flag-EMC7 cells); and Flag-EMC6 was expressed at slightly reduced levels (compare with the endogenous EMC6 level in WT cells). Note that we did not immunoblot the Flag-EMC6 elution fractions, which were concentrated by sedimentation and used in their entirety for sequencing.

**Extended Data Fig. 2 | Quality control for the selective ribosome profiling.**

(a) Distribution of footprint lengths in coding regions of mRNAs. (b) Reads were grouped based on read lengths (18–35 nt), and the fraction of reads assigned to the three different frames is shown. Strong 3-nt periodicity of read distribution was observed for most footprint sizes. (c) Ribosomal A-site adjusted read distribution around the start and stop codons of mRNAs. (d) Scatterplots show

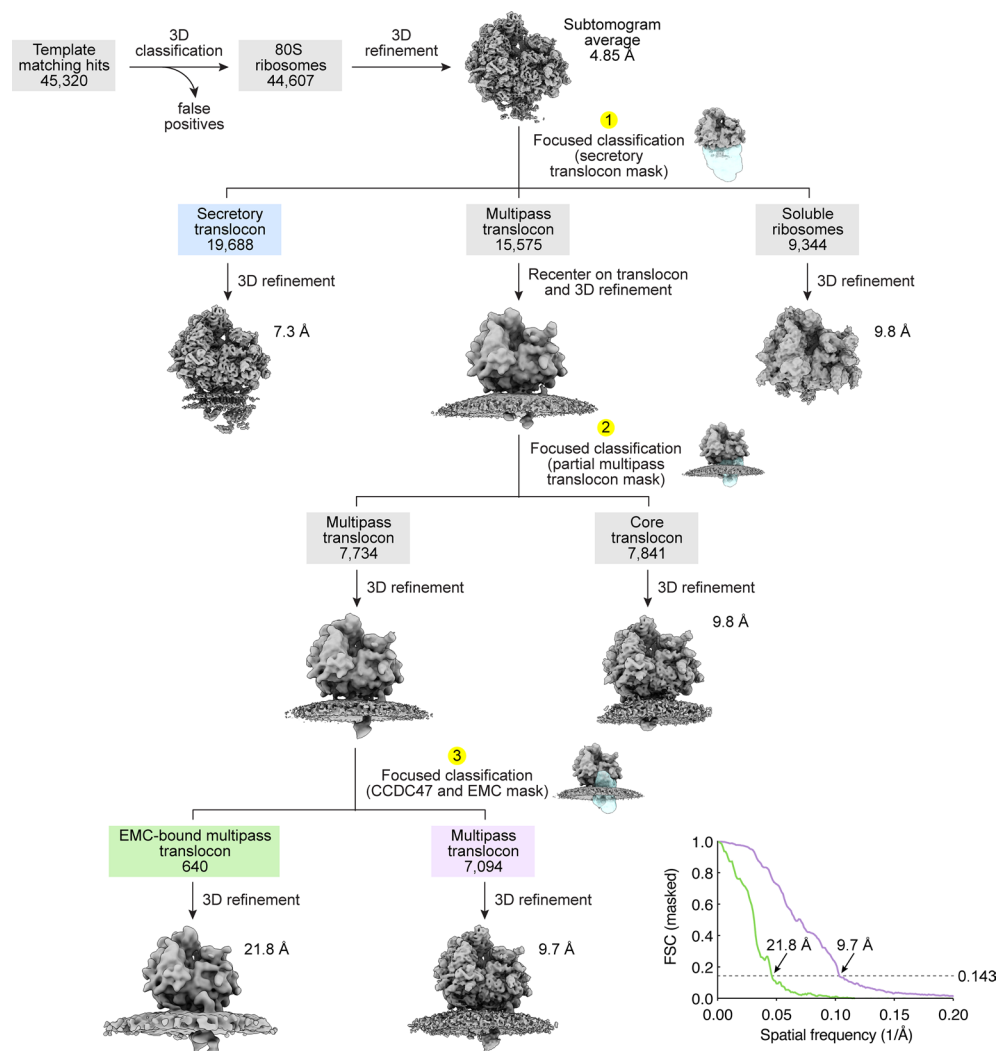
the correlation of gene-level enrichment ($\log_2(\text{IP}/\text{Input})$) between biological replicates for each subunit. Each point represents one gene, and all expressed coding genes are included in the analyses. Pearson correlation coefficient values are indicated. (e) Scatterplots show the correlation of gene-level enrichment ($\log_2(\text{IP}/\text{Input})$) between different EMC subunits.



Extended Data Fig. 3 | Additional characterization of EMC recruitment.

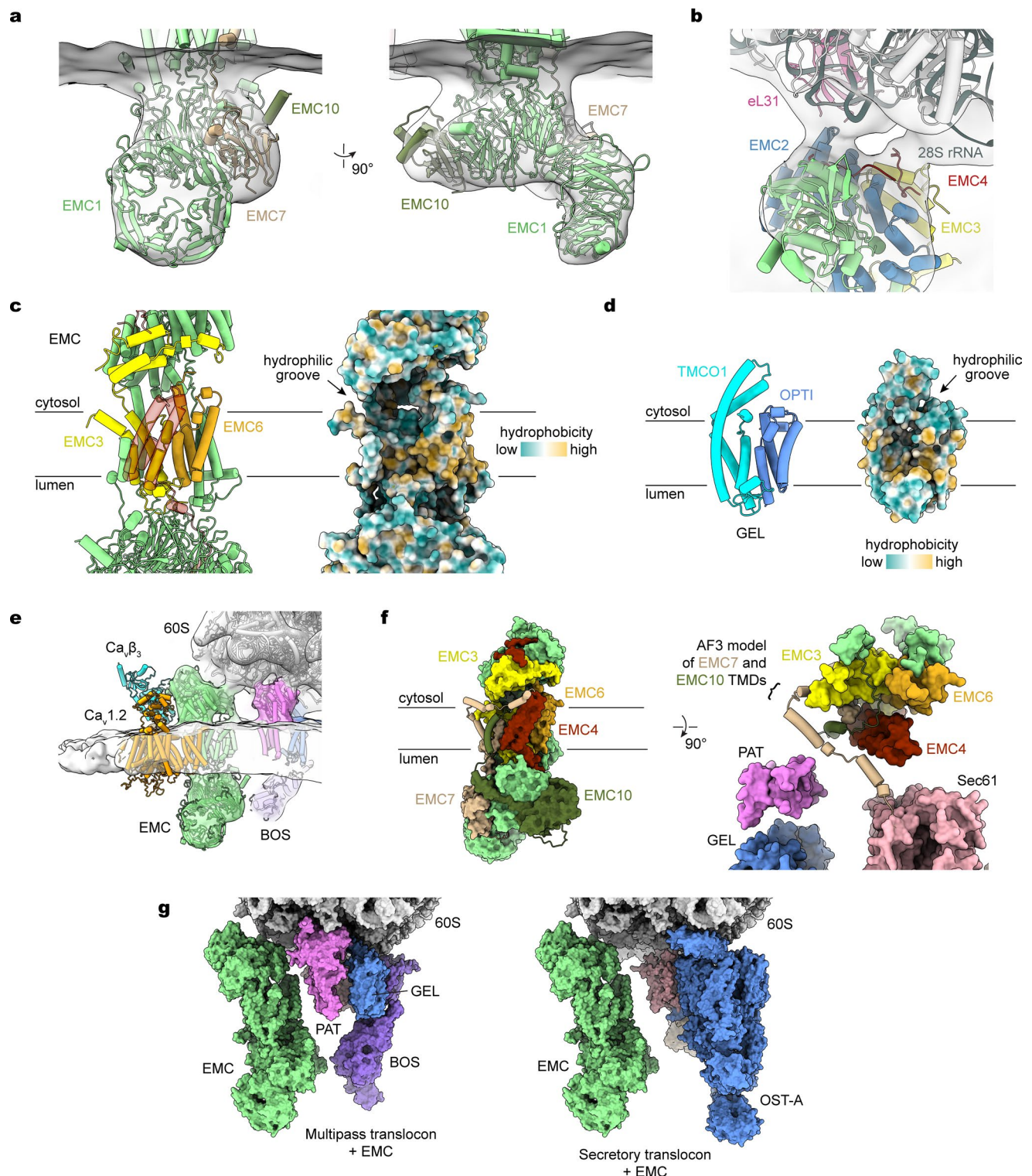
(a) Stalled Flag-tagged ABCD4 constructs truncated 45 residues downstream of TMD4 (4-TMD) were translated in rabbit reticulocyte lysate in the presence of microsomes derived from Twin-Strep-tagged CCDC47 or OST48 HEK293 cells. After isolating the membrane-associated fraction by sedimentation, digitonin-solubilized membranes were subjected to tandem affinity purification via the Flag- and TwinStrep tags, and analyzed by SDS-PAGE and western blotting. **(b)** EMC and MPT interaction profiles for ANO5, a 10-TMD type II protein with a 301-residue cytosolic N-tail, which engaged MPT and EMC early, after its first

(closely-spaced) TMD pair emerged from the ribosome. Mean enrichment (line) and standard deviation (shaded) are shown. **(c)** Interaction profiles for SLC30A5, a 16-TMD type II protein with short cytosolic segments that engaged MPT early, but did not recruit EMC until TMD6 had emerged from the ribosome. **(d)** Interaction profiles for ATG9A, a 4-TMD type II protein with several long cytosolic segments, that engages both EMC and MPT. **(e)** Interaction profiles for TMEM205, a 4-TMD type III protein with short cytosolic segments that recruits MPT but not EMC.



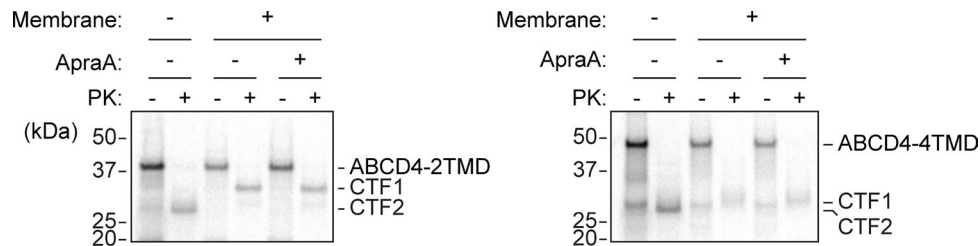
Extended Data Fig. 4 | Cryo-ET data processing scheme. Template matching with pytom-match-pick was used to identify 45,320 ribosome particle candidates, which were extracted in Warp and analyzed in RELION. Initial coarse 3D classification without image alignment was used to remove false positives from the dataset. The remaining 44,607 particles were then subject to several rounds of focused classification using a mask around the secretory translocon (mask 1) to distinguish between soluble ribosomes and ribosomes bound to secretory- or multipass translocons. Multipass translocon particles were

recentered on Sec61 and after refinement, were subject to 3D classification focused on Sec61 and portions of the BOS (Nicalin and TMEM147) and PAT (CCDC47) complexes (mask 2) to remove core translocon variants. The resulting 7,734 multipass translocon particles were subject to an additional round of 3D classification focused on CCDC47 and weak density corresponding to EMC (mask 3). This yielded two classes corresponding to the multipass translocon with and without EMC. Final classes were refined in M. Fourier shell correlation (FSC) curves for the two maps are shown.



Extended Data Fig. 5 | Additional structural analysis. (a) Orthogonal views of the human EMC (PDB ID 9C7V) luminal domain after docking as a single rigid-body into the cryo-ET map, globally filtered to 21.8 Å. (b) Close-up view of the EMC-ribosome contact region. (c) View into the hydrophilic groove formed by the conserved Oxa1 superfamily insertase module of EMC (EMC3, yellow; EMC6, goldenrod; EMC4, transparent maroon). (d) View into the hydrophilic groove formed by the conserved Oxa1 superfamily insertase module of MPT (TMCO1, cyan; OPT1, cornflower blue). (e) Model for assembly of the cardiac voltage-gated calcium channel $\text{Ca}_v1.2$ at the EMC-bound multipass translocon, generated by superimposing the previously determined EMC- $\text{Ca}_v1.2$ - $\text{Ca}_v\beta_3$ structure (PDB ID 8EOI) onto the structure of ribosome-bound EMC. (f) Orthogonal views of an

AlphaFold3 model of human EMC superimposed onto a surface representation of the EMC-bound multipass translocon; only EMC is shown in the left panel for clarity. The flexible C-terminal regions of EMC7 (tan) and EMC10 (dark olive) from the AlphaFold3 model are shown in cartoon representation to highlight their approximate positions near EMC3 (yellow) and EMC4 (maroon) on the insertase side of EMC, in good agreement with the site-specific photocrosslinking experiments. (g) Side-by-side comparison of the EMC binding mode at the multipass- (left) and secretory (right) translocons. The model at right was obtained by superimposing EMC, Sec61 and the 60S ribosomal subunit as a single rigid body onto a previously determined cryo-ET model of the secretory translocon (PDB ID 8B6L) via the C-terminal half of Sec61A1.



Extended Data Fig. 6 | The first two TMD pairs of ABCD4 insert independently of the Sec61 lateral gate. ³⁵S-methionine labeled ABCD4 2-TMD (left) and 4-TMD (right) constructs were translated in rabbit reticulocyte lysate in the presence or absence of HEK293-derived membranes and the Sec61 lateral

gate inhibitor, Aprataxin A (ApraA), isolated by sedimentation, and analyzed by autoradiography before or after proteinase K (PK) treatment. C-terminal protected fragments (CTF1 and CTF2) resulting from PK digestion are indicated.

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| ☐ | ☒ | A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
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| ☐ | ☒ | A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ | For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
<i>Give P values as exact values whenever suitable.</i> |
| ☒ | ☐ | For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ | For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ | Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated |

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	The following software was used for cryo-ET data collection and processing: SerialEM, Warp-2.0.0, IMOD-4.12.52, pytom-match-pick-0.11.0 RELION-5.0.1, M-2.0.0
Data analysis	The following software was used to analyze the Rfoot-seq data: Bowtie v2.2.6; TopHat v2.1.0; HTSeq-count v2.0.3; python 3.10; samtools v1.9; deeptools v3.1.1; DeepTMHMM; RibORF 2.0

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Raw sequencing data are available at the NCBI Gene Expression Omnibus repository with the accession code GSE319640. Cryo-ET maps are available at the Electron

Microscopy Data Bank with accession codes: EMD-75661 (ribosome-bound multipass translocon) and EMD-75658 (ribosome-bound multipass translocon with EMC). All other data are available in the main text or the supplementary materials.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender N/A

Reporting on race, ethnicity, or other socially relevant groupings N/A

Population characteristics N/A

Recruitment N/A

Ethics oversight N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size Sample size for each analysis is indicated in figure legends and methods.

Data exclusions No data were excluded from the analysis.

Replication Footprinting analysis from cells was performed in two biological replicates for Flag-EMC6 and Flag-EMC7, and three biological replicates for Flag-EMC10. Except where noted, the Flag-EMC6 Flag-EMC7 and Flag-EMC10 samples were combined into a single sample ("EMC") (n=7). The cryo-ET reconstructions were obtained by combining five datasets from four independent samples (44,607 total particles). All in vitro analyses are representative of at least two independent experiments performed in part or in full with similar results.

Randomization N/A

Blinding N/A

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

- n/a Involved in the study
- ☐ ☒ Antibodies
 - ☐ ☒ Eukaryotic cell lines
 - ☒ ☐ Palaeontology and archaeology
 - ☒ ☐ Animals and other organisms
 - ☒ ☐ Clinical data
 - ☒ ☐ Dual use research of concern
 - ☒ ☐ Plants

Methods

- n/a Involved in the study
- ☒ ☐ ChIP-seq
 - ☒ ☐ Flow cytometry
 - ☒ ☐ MRI-based neuroimaging

Antibodies

Antibodies used	The following antibodies were used for immunoblotting. Except as noted, antibodies were used at 1:1,000 dilution. Antibodies against human TMCO1, Sec61β (1:10,000) and TRAPα (1:5,000) were characterized previously. Other antibodies were obtained from the following commercial sources: rabbit anti-CCDC47 (A305-100A), rabbit anti-EMC1 (A305-605A) antibodies from Bethyl Laboratories; goat anti-NOMO1 (PA5-47534) from Thermo Fisher; rabbit anti-mouse IgG H&L (HRP) (ab6728; 1:5000), rabbit anti-EMC10 (ab180148), and rabbit anti-uL2 (ab169538) antibodies from Abcam; mouse anti-Flag (F1804; 1:2,000), donkey anti-rabbit IgG-peroxidase (SAB3700863; 1:5000), and rabbit anti-goat IgG-peroxidase (A5420; 1:5000) antibodies from MilliporeSigma; rabbit anti-EMC4 (27708-1-AP) and rabbit anti-EMC7 (27550-1-AP) antibodies from Proteintech; rabbit anti-Sec61A1 (14868) and rabbit anti-Sec61β (14648; 1:3,000) antibodies from Cell Signaling Technology; rabbit anti-uL22 antibody (AP9892b) from Abgent; mouse anti-OST48 (sc-74408) antibody from Santa Cruz Biotechnology; mouse anti-STT3A (H00003703-M02) antibody from Novus Biologicals; rabbit anti-EMC6 (ARP44679_P050) antibody from Aviva Systems Biology. The following antibodies were used for immunoprecipitation: anti-Sec61A1 (Cell Signaling Technology 14868), anti-Asterix (Thermo Fisher PA5-66788), anti-TMCO1 (Abcam ab220729), anti-EMC2 (Proteintech 25443-1-AP), anti-EMC3 (Thermo Fisher 711771), anti-EMC4 (Bethyl A305-752A-M), anti-EMC7 (Bethyl A305-678A-M) and anti-EMC10 (Abcam ab180148).
Validation	Antibodies were validated for specificity against the human antigen by the manufacturer or in earlier published work as follows: anti-TMCO1 (PMID:41116047), anti-CCDC47 (Bethyl Laboratories, PMID:41116047), anti-EMC1 (Bethyl Laboratories, PMID:32459176), anti-EMC4 (Bethyl Laboratories, PMID:32459176), anti-EMC7 (Bethyl Laboratories, PMID:32459176), anti-NOMO1 (Thermo Fisher, PMID:36261522), anti-EMC3 (Thermo Fisher, PMID:37957425), anti-Asterix (Thermo Fisher, PMID:36261522), anti-EMC10 (Abcam, PMID: 33236988), anti-EMC4 (Proteintech, PMID:37199759), anti-EMC7 (Proteintech, PMID:37199759), anti-EMC2 (Proteintech, PMID:37199759), anti-uL22 (Abgent, PMID:41116047), anti-OST48 (Santa Cruz Biotechnology, PMID:41116047) and anti-STT3A (Novus Biologicals, PMID:41116047). Anti-EMC7 (Proteintech) and anti-EMC10 (Abcam) were also validated in experiments using tagged cell lines, as described in this manuscript.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)	Flp-In T-REx 293 cell line are from Invitrogen
Authentication	Flp-In T-REx 293 cell line was authenticated by the antibiotic resistance markers within its genome. EMC6-KO cells were generated and validated as in PMID 29242231.
Mycoplasma contamination	Cells were checked approximately every six months for mycoplasma contamination using the Universal Mycoplasma Detection kit (ATCC), and were found to be negative.
Commonly misidentified lines (See ICLAC register)	None used

Plants

Seed stocks	N/A
Novel plant genotypes	N/A
Authentication	N/A